INCH-POUND

MIL-M-14/6
2 April 1990

MILITARY SPECIFICATION SHEET

MOLDING COMPOUNDS, DIALLYL ORTHO-PHTHALATE, THERMOSETTING, CONTAINING MINERAL FILLERS, GLASS OR POLYMERIC FILLERS

This specification is approved for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-M-14.

Diallyl orth-phthalate resin molding compounds, shall be of the following types, as specified:

Type GDI-30	–	Long glass fiber filler, general-purpose, nominal-impact strength, 3.0 foot-pounds per inch notch
Type GDI-30F	–	Long glass fiber filler, flame resistant. nominal-impact strength, 3.0 foot pounds per inch notch
Type GDI-300	–	Long glass fiber filler, general-purpose, nominal-impact strength, 30.0 foot-pounds per inch notch
Type GDI-300F	–	Long glass fiber filler, flame-resistant, nominal-impact strength, 30.0 foot-pounds per inch notch
Type MDG	–	Mineral filler, general-purpose
Type MDG-F	–	Mineral filler, general-purpose flame resistant
Type SDG	–	Short glass fiber filler, general-purpose
Type SDG-F	–	Short glass fiber filler, flame-resistant
Type SDI-5	–	Acrylic polymer fiber filler, impact-resistant; nominal-impact strength, 0.50 foot-pound per inch notch
Type SDI-30	–	Polyethylene terephthalate fiber filler, impact-resistant; nominal-impact strength, 3.0 foot-pounds per inch notch.

AMSC N/A FSC 9330

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REQUIREMENTS

Qualification test requirements are specified in table I. Batch acceptant test requirements are specified in table II.

NOTES

Type GDI-30. This type is a glass fiber-filled diallyl ortho-phthalate resin molding compound of low-loss, high dielectric strength, low shrinkage, excellent moisture resistance, and relatively high-impact strength.

Type GDI-30F. This type is the same as GDI-30 but is also flame-resistant.

Type GDI-300. This type is a glass fiber-filled diallyl ortho-phthalate resin molding compound of low-loss, high-dielectric strength, low-shrinkage, moisture resistance, and very high-impact strength.

Type GDI-300F. This type is the same as GDI-300 but is also flame-resistant.

Type MDG. This type is a mineral-filled diallyl ortho-phthalate compound for use where good dielectric properties are required.

Type MDG-F. This type is the same as MDG but is also flame-resistant.

Type SDG. This type is a glass-filled diallyl ortho-phthalate resin compound of low-loss, high-dielectric strength, low shrinkage, and good moisture resistance, and relatively low-impact strength.

Type SDG-F. This type is the same as SDG but is also flame-resistant.

Type SDI-5. This type is an acrylic polymer fiber-filled diallyl ortho-phthalate resin compound of low-loss, high-dielectric strength, good moisture resistance, and moderate-impact strength.

Type SDI-30. This type is a polyethylene terephthalate fiber-filled diallyl ortho-phthalate resin compound of low-loss, high-dielectric strength, good moisture resistance, and high-impact strength.

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TABLE I. *Qualification test requirements for diallyl ortho-phthalate resin molding compounds, mineral, glass or polymer fiber filled.*

Requirement	Type GDI-30	Type GDI-30F	Type GDI-300	Type DGI-300F	Type MDG	Type MDG-F	Type SDG	Type SDG-F	Type SDI-5	Type SDI-30
					Mechanical/physical					
Compressive strength, endwise	24,000	24,000	20,000	20,000	20,000	20,000	24,000	24,000	18,000	16,000
Dimensional stability	0.1	0.1	0.1	0.1	0.2	0.2	0.2	0.2	0.2	0.2
Flexural strength	13,000	13,000	35,000	35,000	10,000	10,000	13,000	13,000	8,000	10,000
Heat deflection temperature	180	180	200	200	148	148	160	160	100	115
Heat resistance	155	155	155	155	100	100	120	120	80	90
Impact strength, side[1]	3.0	3.0	30	30	0.35	0.35	0.50	0.50	0.60	3.0
Tensile strength	6,000	6,000	18,000	18,000	6,000	6,000	6,000	6,000	3,500	3,600
Water absorption	0.5	0.5	0.5	0.5	0.7	0.7	0.5	0.5	0.5	0.5
					Electrical					
Arc resistant	115	115	115	115	115	115	115	115	100	115
Dielectric breakdown:										
Short-time test (To be recorded as basis for initial voltage in step-by-step test)										
Step-by-step test	45	45	40	40	40	40	45	45	45	45
Short-time test (To be recorded as basis for initial voltage in step-by-step test)										
Step-by-step test	40	40	35	35	40	40	40	40	40	40
Dielectric constant:										
at 1 kilohertz	4.6	4.6	5.0	5.0	5.0	5.0	4.6	4.6	4.1	4.1
	4.7	4.7	5.0	5.0	5.2	5.2	4.7	4.7	4.2	4.2
at 1 megahertz	4.6	4.6	5.0	5.0	5.0	5.0	4.6	4.6	3.8	3.8
	4.7	4.7	5.0	5.0	5.2	5.2	4.7	4.7	3.9	3.9

See footnote at end of table.

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TABLE I. *Qualification test requirements for diallyl ortho-phthalate resin molding compounds, mineral, glass or polymer fiber filled – Continued.*

Requirement	Type GDI-30	Type GDI-30F	Type GDI-300	Type DGI-300F	Type MDG	Type MDG-F	Type SDG	Type SDG-F	Type SDI-5	Type SDI-30
Electrical *(Continued)*										
Dielectric strength:										
Short-time test	325	325	325	325	325	325	325	325	325	325
Step-by-step test	300	300	300	300	300	300	300	300	300	300
Short-time test	325	325	325	325	325	325	325	325	325	325
Step-by-step test	300	300	300	300	300	300	300	300	300	300
Dissipation factor:										
at 1 kilohertz	0.015	0.015	0.020	0.020	0.019	0.019	0.015	0.015	0.025	0.020
	0.018	0.018	0.025	0.025	0.020	0.020	0.015	0.015	0.028	0.025
at 1 megahertz	0.020	0.020	0.025	0.025	0.025	0.025	0.020	0.020	0.035	0.025
	0.023	0.023	0.030	0.030	0.025	0.025	0.023	0.023	0.025	0.025
Surface resistance	5,000	5,000	5,000	5,000	5.0	5.0	10	5,000	100	10
Volume resistance	5,000	5,000	5,000	5,000	2.0	2.0	10	5,000	100	10
Water extract conductance	50×10^{-6}	50×10^{-6}	–	–	–	–	–	–	–	–
Combustion										
Flame resistance:										
Ignition time	–	90	–	90	–	90	–	90	–	–
Burning time	–	90	–	90	–	90	–	90	–	–
Flammability	–	V-0	–	V-0	–	V-0	–	V-0	–	–
Toxicity when heated:										
Carbon dioxide		15,000	–	15,000		15,000		15,000		
Carbon monoxide		1,000	–	1,000		1,000		1,000		
Ammonia		2,500	–	25,000		2,500		2,500		
Aldehydes as HCHO		50	–	50		50		50		
Cyanide as HCN		60	–	60		60		60		
Oxide of nitrogen as NO_2		100	–	100		100		100		
Hydrogen chloride		100	–	100		100		100		

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TABLE II. *Batch acceptance test requirements for diallyl ortho-phthalate molding compounds, mineral, glass or polymer fiber filled.*

Property to be tested	Type GDI-30	Type GDI-30F	Type GDI-300	Type DGI-300F	Type MDG	Type MDG-F	Type SDG	Type SDG-F	Type SDI-5	Type SDI-30
	Performance requirement for each compound.									
Arc resistance	115	115	115	115	115	115	115	115	100	115
Dielectric constant at 1 megahertz	4.7	4.7	5.0	5.0	5.0	5.0	4.7	4.7	3.9	3.9
Dielectric strength, step-by-step	300	300	300	300	300	300	300	300	300	300
Dissipation factor at 1 megahertz	0.023	0.023	0.030	0.030	0.030	0.030	0.023	0.023	0.025	0.025
Flexural strength	13,000	13,000	35,000	35,000	10,000	10,000	13,000	13,000	8,000	10,000
Impact strength, side[1]	3.0	3.0	30.0	30.0	0.30	0.30	0.50	0.50	0.60	3.0
Water absorption	0.5	0.5	0.5	0.5	0.7	0.7	0.5	0.5	0.5	0.5
Water extract conductance	50×10^{-6}	50×10^{-6}	–	–	–	–	–	–	–	–

[1]The side of the test specimen is that area formed by the chase of the mold.

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User activities:
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Preparing activity:
Navy - SH
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